

| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks available |     |     |       |       |      |
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|          |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 7        |  |  |  | <p><b>Indicative content</b></p> <ul style="list-style-type: none"> <li>ions are free to move because aluminium oxide is molten</li> <li>ions travel to the oppositely charged electrodes because opposite charges attract / <math>\text{Al}^{3+}</math> ions go to the cathode / <math>\text{O}^{2-}</math> ions go to the anode</li> <li>reduction is gain of electrons, oxidation is loss of electrons</li> </ul> <p>at the cathode – reduction</p> <ul style="list-style-type: none"> <li><math>\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}</math></li> <li>aluminium ions gain 3 electrons to form aluminium atoms</li> </ul> <p>at the anode – oxidation</p> <ul style="list-style-type: none"> <li><math>2\text{O}^{2-} \rightarrow \text{O}_2 + 4\text{e}^-</math></li> <li>two oxide ions lose 2 electrons each to form two oxygen atoms</li> <li>these pair up to form oxygen molecules</li> </ul> <p><b>5-6 marks</b><br/>           Good explanation of the movement of ions towards both electrodes and of reduction and oxidation of both ions in terms of gain/loss of electrons; attempt at electrode equations<br/> <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i></p> | 6               |     |     | 6     |       |      |